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Action-Angle Clustering of Stellar Streams in Merging Galaxies

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ABSTRACT

Stellar streams are sensitive tracers of the Galactic gravitational field and provide a promising probe of the dark-matter distribution in the Milky Way halo. Action-clustering methods exploit the fact that canonical actions are conserved in a time-independent integrable potential. However, the Milky Way is neither static nor isolated, and merger-induced perturbations may alter both the stream population and the reliability of action-clustering as a potential diagnostic.

We test the robustness of radial-action clustering using simulated globular-cluster streams evolved in live Milky Way-like halos undergoing mergers with different mass ratios and orbital configurations. At each instant, we define a theoretical locally matched spherical reference potential, $\Phi_0(t)$, whose radial force best represents the true acceleration field over the region occupied by a stellar cohort. When $\Phi_0(t)$ is frozen, it assigns every stellar phase-space point a canonical instantaneous reference action, $J_0(t)$. Along the real merger trajectories, these actions evolve through both the residual force not reproduced by $\Phi_0(t)$ and the change of the locally matched reference potential itself.

We identify locally coherent stream cohorts using observable phase-space criteria designed to favor similar instantaneous-action evolution among their members, including smooth along-stream velocity profiles, unfolded radial phase-space structure, limited spatial extent, and locations away from radial turning points. For each selected cohort, we fit a spherical NFW potential by maximizing the one-dimensional Kullback–Leibler divergence of the canonical fitted-model radial actions, $J_{r,\mathrm{fit}}$. We also compare this fitted distribution with the frozen-BFE radial-cycle proxy distribution, $\widetilde{J}_{r,\mathrm{BFE}}$, evaluated from the same stellar phase-space points. The latter characterizes radial orbital structure in the frozen non-integrable potential and need not be a conserved canonical action.

At the distance-matched fiducial 1:10 merger snapshot, the selected cohorts provide a useful recovery of the inner-halo enclosed-mass profile. Across the time evolution, stronger mergers remove usable coherent material earlier and produce larger and more rapidly varying mass biases, whereas the 1:100 merger remains comparatively stable. We identify three limitations: loss of coherent tracer material, disagreement between the fitted canonical-action distribution and the frozen-BFE radial cycle proxy distribution, and a local-to-global mismatch caused by halo non-sphericity and spatial sampling. While usable cohorts remain, the largest fit-to-local mass errors are generally accompanied by the largest fitted-versus-proxy discrepancies. By contrast, the local-to-global contribution remains mostly below ten percent, even for the 1:2 merger, over the analyzed BFE-valid intervals. Observable cohort selection can therefore improve the robustness of action-clustering fits, but it cannot guarantee unbiased potential recovery in a strongly perturbed halo.

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1. INTRODUCTION

Stellar streams form through the tidal disruption of globular clusters and dwarf galaxies. Because their stars originate from dynamically cold progenitors and subsequently follow neighboring trajectories in the Galactic gravitational field, the debris forms thin, coherent structures in phase space. Although a stream may extend over a large spatial region, its small internal phase-space dispersion allows it to retain information about the mass profile, shape, and dynamical evolution of the Milky Way halo.

One way to extract this information is through the distribution of stream stars in action space. In a time-independent integrable Hamiltonian, canonical actions label invariant orbital tori and remain constant along each orbit. Stars stripped from a dynamically cold progenitor are therefore expected to occupy neighboring regions of action space when their phase-space coordinates are evaluated in an appropriate integrable potential. This motivates potential-fitting methods that search for the potential model in which stream actions are maximally clustered. R. E. Sanderson et al. (2015)

formulated this approach using the Kullback–Leibler divergence (KLD), and subsequent studies have applied and tested it using mock and observed stream samples (e.g., S. Reino et al. 2021, 2022). More generally, the construction and numerical accuracy of action coordinates depend on the integrability or approximate separability of the adopted potential (J. L. Sanders & J. Binney 2016).

Action clustering is nevertheless not an identity between maximum compactness and the correct potential. An incorrect trial potential can shift an action distribution without substantially broadening it, and a short stream segment with limited orbital-phase coverage may not distinguish between competing models. Potential recovery therefore depends not only on the intrinsic coldness of the tracer, but also on its coverage in orbital phase, radius, energy, and angular momentum. The orbital-phase dependence found by S. Reino et al. (2021), together with the improved constraints obtained by combining dynamically distinct streams, illustrates the importance of this sampling problem.

A live merger creates a more fundamental theoretical complication. The actual Galactic Hamiltonian is time dependent and generally non-integrable, so a global set of conserved canonical actions for the full system is not guaranteed to exist. To define a local spherical dynamical target, we introduce at each instant a theoretical reference potential, $\Phi_0(t)$, whose radial force best represents the true acceleration field over the region occupied by a stellar cohort. When $\Phi_0(t)$ is frozen, each actual stellar phase-space point can be assigned to an invariant torus of the reference Hamiltonian and therefore given a canonical instantaneous reference action, $J_0(t)$. As the merger evolves, both the stellar phase-space point and the locally matched reference potential change. The resulting evolution of J_0 contains a physical-response contribution from the actual-minus-reference force and a reference-evolution contribution from the sensitivity of the action to the changing $\Phi_0(t)$. A nearly common change among the members can translate the reference-action cloud without strongly broadening it, whereas member-dependent changes diffuse or distort the cloud. This motivates selecting cohorts whose present-day phase-space structure favors similar instantaneous-action evolution before applying the KLD potential fit.

The Milky Way provides an important example of this non-equilibrium problem. Its halo is neither isolated nor exactly spherical, and it is perturbed by massive satellites and recent accretion events. In particular, the Large Magellanic Cloud (LMC) can generate a dynamical-friction wake and a large-scale response in the Milky Way halo (C. Vera-Ciro & A. Helmi 2013; N. Garavito-Camargo et al. 2019). Its infall also induces reflex motion between the inner Galaxy and the outer halo, which can bias dynamical inferences that assume a stationary Galactocentric frame (M. S. Petersen & J. Peñarrubia 2020). Observed and simulated stellar streams exhibit track–velocity misalignments, orbital deflections, and other perturbations associated with the LMC and the deformation of the Milky Way halo (e.g., D. Erkal et al. 2018; N. Shipp et al. 2019; R. A. N. Brooks et al. 2025).

Recent studies have begun to connect these perturbations directly to action-based potential inference. A. Arora et al. (2022) showed that massive mergers can displace or disrupt stream structures in action space and that time-independent potential representations can lose orbital information on relatively short timescales. Using a time-evolving basis-function-expansion model of the Milky Way–LMC system, R. A. N. Brooks et al. (2024) found that radial-action clustering can retain useful local mass information for shorter-period streams without close LMC encounters, whereas longer-period and strongly perturbed streams may fail to recover the spherically averaged mass profile. These results motivate a distinction between the canonical actions calculated in a fitted integrable model and action-like diagnostics constructed from a more realistic, but potentially non-integrable, potential field.

In this work, we study that distinction using simulated globular-cluster streams evolved in live Milky Way-like halos undergoing mergers with different mass ratios and orbital configurations. Rather than using all available stream material, we select local coherent cohorts with smooth along-stream velocity profiles, unfolded radial phase-space structure, limited spatial extent, and locations away from radial turning points. These observable criteria are intended to reject obviously folded, phase-mixed, or kinematically irregular material. They do not, by themselves, demonstrate that the retained stars guarantee an unbiased potential fit. We have shown that the post-selected stellar cohorts can retain the spherically-averaged mass profile at the effective current snapshot to a reasonable precision.

For every selected cohort, we fit a spherical NFW potential by maximizing the one-dimensional KLD of the canonical radial-action distribution, $J_{r,\mathrm{fit}}$. We then freeze the BFE representation of the simulated halo at the same snapshot and integrate each selected star through one radial cycle to construct $\widetilde{J}_{r,\mathrm{BFE}}$. Because the frozen BFE field is generally non-spherical and need not be integrable, $\widetilde{J}_{r,\mathrm{BFE}}$ is treated as a radial-cycle action proxy rather than as a true instantaneous canonical action. We compare the fitted and proxy distributions through their star-count-weighted median displacement and their difference in radial-action KLD, with both KLD values evaluated against the same fixed reference distribution. These comparisons are simulation-based validation diagnostics of how well the optimized spherical model reproduces the radial orbital structure sampled in the frozen potential field.

We further decompose the enclosed-mass bias into two components. The fit-to-local component measures whether the KLD-optimized spherical NFW model reproduces the spherical-equivalent radial force sampled locally by the selected cohorts. The local-to-global component measures whether that locally sampled spherical-equivalent radial force is representative of the globally spherically averaged BFE mass profile. This decomposition separates failure of the fitted spherical model from the additional bias produced by non-sphericity and non-uniform spatial sampling.

Our results show that stronger mergers generally remove usable coherent material earlier and produce larger and more rapidly varying mass biases, whereas weaker mergers preserve useful constraints for longer. We identify three corresponding limitations: disappearance of a usable tracer population; mismatch between the canonical fitted-model radial-action distribution and the frozen-BFE radial-cycle proxy distribution; and disagreement between the locally sampled force field and the global spherical average. While usable cohorts remain, large fit-to-local mass errors generally coincide with large discrepancies between fitted action versus action-proxy. The local-to-global contribution is typically smaller and remains mostly below ten percent over the analyzed BFE-valid intervals, even for the 1:2 merger. The relationships between the scalar action diagnostics and mass bias are nevertheless descriptive rather than universal, particularly in the most strongly perturbed case, where changes in distribution shape and orbital sampling cannot be represented by a single displacement or clustering statistic.

Section 2 develops the canonical-action, reference-action, and radial-cycle-proxy framework. Section 3 presents the observable cohort-selection criteria. Section 4 describes the simulations, examines the time-dependent mass recovery, and analyzes the three failure modes. Section 5 discusses the dependence on perturber orbital configuration, the limitations of observable cohort selection, and directions for future work.

2. THEORETICAL FRAMEWORK

2.1. *Canonical Actions and Potential-dependent Action Clustering*

For a Hamiltonian system with n degrees of freedom,the canonical action variables are well defined for regular motion confined to invariant tori. In particular, Liouville integrability requires n functionally independent integrals of motion F_i that are mutually in involution,

$$\{F_i, F_j\} = 0 \qquad \text{for all } i, j. \tag{1}$$

When the corresponding common level set is regular, compact, and connected, it forms an invariant n-torus. The canonical actions are then defined by integrals over the n independent fundamental cycles γ_i of this torus:

$$J_i = \frac{1}{2\pi} \oint_{\gamma_i} \boldsymbol{p} \cdot d\boldsymbol{q}. \tag{2}$$

In separable canonical coordinates, this expression reduces to

$$J_i = \frac{1}{2\pi} \oint_{\gamma_i} p_i \, dq_i. \tag{3}$$

Here, q_i and p_i are canonically conjugate variables. The actions label the invariant torus and remain constant under the integrable Hamiltonian, while the conjugate angles evolve along it.

A static spherically symmetric potential is integrable. Away from degenerate orbits, the energy H, total angular momentum squared L^2, and azimuthal angular momentum L_z provide three functionally independent integrals in involution. Consequently, three canonical actions (J_r, J_θ, J_ϕ) can be defined. Throughout the following expressions, momenta and energies are taken per unit stellar mass. The energy is

$$\begin{aligned} E &= \frac{1}{2}\left(p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\phi^2}{r^2 \sin^2\theta}\right) + \Phi(r) \\ &= \frac{1}{2}\left(p_r^2 + \frac{L^2}{r^2}\right) + \Phi(r), \end{aligned} \tag{4}$$

where

$$L^2 = p_\theta^2 + \frac{p_\phi^2}{\sin^2\theta}. \tag{5}$$

Since the azimuthal angle ϕ is cyclic in a spherical potential, its conjugate momentum is conserved:

$$p_\phi = L_z. \tag{6}$$

So that the azimuthal action is:

$$J_\phi = \frac{1}{2\pi} \oint_l p_\phi \cdot d\phi = p_\phi = L_z \tag{7}$$

Using the expression of angular momentum shown in Equ.4, the polar action can be expressed as:

$$J_\theta = \frac{1}{2\pi} \oint_l p_\theta \cdot d\theta = \frac{1}{\pi} \int_{\theta_{min}}^{\theta_{max}} \sqrt{L^2 - \frac{L_z^2}{\sin^2\theta}} \cdot d\theta \tag{8}$$

Using $\sin(\theta_{min}) = \sin(\theta_{max}) = \frac{|L_z|}{L}$, the integral gives:

$$J_\theta = L - |L_z| \tag{9}$$

Finally, using Equ.4, the radial action can also be acquired:

$$J_r = \frac{1}{\pi} \int_{r_{min}}^{r_{max}} \sqrt{2[E - V(r)] - \frac{L^2}{r^2}} \cdot dr \tag{10}$$

where $r_{\rm min}$ and $r_{\rm max}$ are the radial turning points in the adopted spherical potential. For the fitted static spherical halo models, we calculate these canonical actions using `Agama`. The code determines the allowed radial interval and evaluates the corresponding canonical actions.

Stars stripped from a dynamically cold progenitor begin on nearby orbits and therefore occupy a relatively compact region in action space when evaluated in an appropriate integrable potential. This compactness contains information about the potential. Stars from the same stream generally sample different orbital phases. *An appropriate potential maps these different phase-space points onto nearby invariant tori,* whereas an incorrect trial potential generally assigns phase-dependent action errors, broadening or distorting the inferred action cloud. This is the physical basis of action-clustering potential inference (R. E. Sanderson et al. 2015; S. Reino et al. 2021, 2022).

This statement is not an identity: a wrong potential may translate an action cloud without greatly broadening it, and a short stream segment may provide insufficient phase coverage to distinguish trial models. Potential recovery therefore requires both a dynamically cold tracer and sufficient diversity in orbital phase, radius, energy, or angular momentum. Combining multiple streams on different orbits can reduce individual-stream degeneracies and biases (R. E. Sanderson et al. 2015; S. Reino et al. 2021).

2.2. *1D Kullback-Leibler Divergence*

We quantify radial-action clustering using the Kullback–Leibler divergence (KLD). For a radial-action probability density $p(J_r)$ and a reference density $q(J_r)$,

$$D_{\mathrm{KL}}[p\|q] = \int p(J_r) \ln\left[\frac{p(J_r)}{q(J_r)}\right] dJ_r. \qquad (11)$$

where $p(J_r)$ is the probability density of the radial action distribution at the points J_r, and $q(J_r)$ is the background distribution.

Also, following the work by R. E. Sanderson et al. (2015), I use the Modified Breiman Density Estimator to calculate the density profile of the action points. According to B. J. Ferdosi et al. (2011), a fine grid will be built across the action space, and the density value at the center of each fine grid $J_{r,k}$ will be evaluated using the following formula:

$$p(J_{r,k}) = \frac{1}{N}\Sigma_{i=1}^{N}(\sigma\lambda_i)^{-d} K_E\left(\frac{J_{r,k}-J_{r,i}}{\sigma_r\lambda_i}\right) \qquad (12)$$

Here, i is the star index, N stands for the total number of stars, and d stands for the dimensionality of the action space. This estimator uses the Epanechnikov Kernel K_E instead of a Gaussian Kernel:

$$K_E(t) = \begin{cases} \frac{d+2}{2V_d}(1-t^2), & \text{if } t^2 < 1, \\ 0, & \text{otherwise.} \end{cases} \qquad (13)$$

, where V_d is the volume of the unit sphere in d-dimensional space. We can easily see the the kernel contribution decreases with normalized distance from the sample point to the grid center, which is as expected. Following the analysis in B. J. Ferdosi et al. (2011), the window sizes σ_j in each direction in the action space is derived from coordinate percentiles in each dimension in the action space:

$$\sigma_r = \min\left\{\frac{\mathrm{P}_{80}(J_{r,i}) - \mathrm{P}_{20}(J_{r,i})}{\ln \mathrm{N}}\right\} \qquad (14)$$

, where $\mathrm{P}_n(J_{r,i})$ is the nth percentile of the radial action values. In addition, define a pilot density $\mathrm{p}_{\mathrm{pilot}}(J_{r,i})$ to describe the degree of clustering for the data point by using Equ. 12, with $\lambda_i = 1$. And the adaptive bandwidth λ_i of this estimator is designed to be inversely proportional to the $\mathrm{p}_{\mathrm{pilot}}(J_{r,i})$:

$$\lambda_i = \left(\frac{\mathrm{p}_{\mathrm{pilot}}(J_{r,i})}{g}\right)^{-\alpha} \qquad (15)$$

, where g is the geometric mean of the pilot density:

$$g = \exp\left[\ \frac{1}{N}\ \Sigma_{i=1}^{N} \ln\ \mathrm{p}_{\mathrm{pilot}}(J_{r,i})\ \right] \qquad (16)$$

and $\alpha = \frac{1}{d}$ is the adaptive-bandwidth exponent. This adaptive bandwidth shrinks kernels in dense regions and widens them in sparse regions, thus increasing the density value where the action points cluster tightly.

Also, for those grids with overly small $p(J_r)$ values, a floor value (like 10^{-9}) needs to be assigned to prevent error. Luckily, the function "`modified_breiman_density`" in the "`MBEtree`" package can realize the above calculation for $p(J_r)$.

Following the action-clustering framework of R. E. Sanderson et al. (2015) and its applications by S. Reino et al. (2021, 2022), trial gravitational potentials are compared through the degree of clustering of the actions calculated from the same observed phase-space data. In this work, the reference distribution $q(J_r)$ is uniform over one fixed and sufficiently wide action interval $[J_{r,\min}, J_{r,\max}]$ that is shared by every trial potential,

$$q(J_r) = \frac{1}{J_{r,\max} - J_{r,\min}}. \qquad (17)$$

The interval $[J_{r,\min}, J_{r,\max}]$ is kept the same for all trial potentials. In practice, we choose this universal range to be sufficiently wide to contain the radial actions produced over the full set of potentials considered, including deliberately extreme choices of the potential parameters. This ensures that changes in the KLD are not driven by changing the support of the background distribution from one potential to another.

Thus, the KLD measures how strongly the radial-action distribution departs from a uniform distribution over the fixed reference range. Since the reference interval is held fixed for all trial potentials, maximizing $D^I_{\rm KL}$ is equivalent to finding the potential in which the stream is most concentrated in J_r, or equivalently, has the lowest radial-action entropy. A mismatched potential generally broadens the inferred J_r distribution of a stream, thereby reducing the KLD. The KLD test used here is based on the expectation that the most accurate potential produces the sharpest radial-action clustering relative to the same uniform background. The success of this method has been demonstrated in static test potentials and in applications combining multiple stellar streams (R. E. Sanderson et al. 2015; S. Reino et al. 2021, 2022).

2.3. *Evolution of Instantaneous Actions in a Locally Matched Spherical Reference Potential*

A live merger creates a conceptually different problem from the static integrable case. The actual Hamiltonian,

$$H(\boldsymbol{z},t) = \frac{1}{2}v^2 + \Phi(\boldsymbol{x},t), \tag{18}$$

is time dependent and generally non-integrable. Consequently, the actual stellar trajectories need not lie on invariant three-tori, and a global set of conserved canonical actions for the merger system is not guaranteed to exist.

Nevertheless, at every instant we may introduce a theoretical spherical and integrable reference potential,

$$\Phi_0(r,t) = \Phi_{\rm sph}[r;\boldsymbol{\lambda}_0(t)]\,, \tag{19}$$

that best represents the local dynamics experienced by a selected stellar cohort. Here, $\boldsymbol{\lambda}_0(t)$ denotes the parameters of the instantaneous reference potential. This theoretical reference potential is distinct from the potential subsequently obtained by KLD maximization. It defines the local spherical target that the KLD procedure will later attempt to recover.

More precisely, let $C(t)$ denote the cohort at time t, and let

$$\boldsymbol{g}(\boldsymbol{x},t) = -\nabla\Phi(\boldsymbol{x},t) \tag{20}$$

be the true instantaneous acceleration field. A locally matched spherical reference may be defined as the spherical potential whose radial force most closely represents the true force over the region occupied by the cohort:

$$\boldsymbol{\lambda}_0(t) \equiv \arg\min_{\boldsymbol{\lambda}} \frac{1}{N_{\mathcal{C}}} \sum_{a\in\mathcal{C}} \left\|\boldsymbol{g}[\boldsymbol{x}_a(t),t] - \boldsymbol{g}_0[r_a(t);\boldsymbol{\lambda}]\right\|^2 . \tag{21}$$

where

$$\boldsymbol{g}_0(r;\boldsymbol{\lambda}) = -\frac{\partial\Phi_{\rm sph}(r;\boldsymbol{\lambda})}{\partial r}\,\hat{\boldsymbol{r}}, \tag{22}$$

and $N_{\mathcal{C}}$ total number of stars in that cohort. Because a spherical model cannot reproduce tangential forces, the relative preference among spherical models in Equation 21 is determined primarily by their ability to reproduce the true local radial acceleration.

The actual Hamiltonian can then be decomposed as

$$H(\boldsymbol{z},t) = H_0[\boldsymbol{z};\boldsymbol{\lambda}_0(t)] + \delta H[\boldsymbol{z},t;\boldsymbol{\lambda}_0(t)]\,, \tag{23}$$

where

$$H_0[\boldsymbol{z};\boldsymbol{\lambda}_0(t)] = \frac{1}{2}v^2 + \Phi_0[r;\boldsymbol{\lambda}_0(t)] \tag{24}$$

is the locally matched spherical reference Hamiltonian, and

$$\delta H \equiv H - H_0 \tag{25}$$

contains the part of the true dynamics not reproduced by that spherical reference.

The time evolution of the theoretical reference can be written schematically as

$$\begin{aligned} \Phi_0(t) &= \mathcal{P}_{\mathcal{C}(t)}[\Phi(t)], \\ \Phi_0(t+dt) &= \mathcal{P}_{\mathcal{C}(t+dt)}[\Phi(t+dt)], \end{aligned} \tag{26}$$

where $\mathcal{P}_{C(t)}$ denotes the locally weighted spherical projection defined by the region occupied by the cohort. The reference potential therefore changes because both the true potential and the locally sampled region evolve. In particular, the residual force at time t changes the subsequent stellar motion and hence the region of the force field occupied by the cohort at $t + dt$.

For a cohort member a, its actual phase-space point at time t can be assigned to an invariant torus of the instantaneous reference Hamiltonian. We define its instantaneous reference action as

$$J^{(a)}_{0,i}(t) \equiv J_i[\boldsymbol{z}_a(t);\boldsymbol{\lambda}_0(t)]\,. \tag{27}$$

This is a canonical action of the instantaneously frozen $H_0(t)$, but it is not generally conserved along the star's real trajectory.

The definition in Equation 27 produces a time-ordered sequence,

$$\begin{aligned} J^{(a)}_{0,i}(t) &= J_i[\boldsymbol{z}_a(t);\boldsymbol{\lambda}_0(t)]\,, \\ J^{(a)}_{0,i}(t+dt) &= J_i[\boldsymbol{z}_a(t+dt);\boldsymbol{\lambda}_0(t+dt)]\,. \end{aligned} \tag{28}$$

Both arguments change between the two instants: the true force moves the star to a new phase-space point, while the locally appropriate spherical reference evolves from $H_0(t)$ to $H_0(t + dt)$.

Taking the total derivative of Equation 27 along the actual trajectory gives

$$\begin{aligned} \frac{dJ^{(a)}_{0,i}}{dt} &= \left\{J^{(a)}_{0,i}, H\right\} + \dot{\lambda}^{\alpha}_0 \left.\frac{\partial J^{(a)}_{0,i}}{\partial\lambda^{\alpha}_0}\right|_{\boldsymbol{z}_a} \\ &= -\left.\frac{\partial\delta H}{\partial\theta_{0,i}}\right|_{\boldsymbol{z}_a} + \dot{\lambda}^{\alpha}_0 \left.\frac{\partial J^{(a)}_{0,i}}{\partial\lambda^{\alpha}_0}\right|_{\boldsymbol{z}_a}, \end{aligned} \tag{29}$$

where repeated reference-potential parameter indices α are summed. The first term is the physical response of the star to the actual-minus-reference Hamiltonian. The second term describes the sensitivity of the assigned action to the evolution of the reference potential itself.

When the residual Hamiltonian is purely a potential difference,

$$\delta H = \delta\Phi = \Phi(\boldsymbol{x}, t) - \Phi_0(r, t), \tag{30}$$

we may define the residual force as

$$\delta\boldsymbol{F} \equiv \boldsymbol{g} - \boldsymbol{g}_0 = -\nabla\delta\Phi. \tag{31}$$

Equation 29 then becomes

$$\frac{dJ_{0,i}^{(a)}}{dt} = \delta\boldsymbol{F}^{(a)} \cdot \left.\frac{\partial\boldsymbol{x}}{\partial\theta_{0,i}}\right|_{\boldsymbol{z}_a} + \dot{\lambda}_0^\alpha \left.\frac{\partial J_{0,i}^{(a)}}{\partial\lambda_0^\alpha}\right|_{\boldsymbol{z}_a}. \tag{32}$$

The first term measures how the part of the true force not reproduced by $\Phi_0(t)$ changes the instantaneous action. The second term measures how the action changes because the spherical reference used to assign it evolves from $\Phi_0(t)$ to $\Phi_0(t+dt)$.

For two cohort members a and b, Equation 32 gives

$$\begin{aligned}\frac{d}{dt}\left(J_{0,i}^{(a)} - J_{0,i}^{(b)}\right) = \Delta_{ab}\left[\delta\boldsymbol{F} \cdot \frac{\partial\boldsymbol{x}}{\partial\theta_{0,i}}\right] \\ + \dot{\lambda}_0^\alpha \Delta_{ab}\left[\left.\frac{\partial J_{0,i}}{\partial\lambda_0^\alpha}\right|_{\boldsymbol{z}}\right],\end{aligned} \tag{33}$$

where

$$\Delta_{ab}X \equiv X^{(a)} - X^{(b)}. \tag{34}$$

If the total right-hand side of Equation 33 remains similar among the cohort members, their pairwise action separations change slowly. The instantaneous-action cloud can then undergo an approximately semi-rigid translation through the time-ordered sequence of reference potentials,

$$\Phi_0(t_1) \to \Phi_0(t_2) \to \cdots \to \Phi_0(t_0), \tag{35}$$

without undergoing strong internal diffusion. By contrast, member-dependent residual forcing or member-dependent sensitivity to the changing reference potential broadens or distorts the instantaneous-action cloud.

This distinction is essential for action-clustering inference. The width of an action cloud evaluated in a trial potential may contain both evolutionary decoherence accumulated before the observed snapshot and additional differential action shifts introduced by a mismatch between the trial potential and the locally appropriate $\Phi_0(t_0)$. KLD maximization can be expected to recover $\Phi_0(t_0)$ only when the first contribution has been sufficiently suppressed. Section 3 therefore introduces observable criteria designed to favor cohorts whose present-day phase-space structure is consistent with an approximately semi-rigid instantaneous-action evolution.

2.4. *Frozen-snapshot Radial-cycle Action-proxy*

On the other hand, if the actual local potential structure differs strongly from being integrable, it will act as an additional factor that declines the accuracy of the KLD potential fitting.

Suppose we know the exact potential model, then at snapshot t_*, we can integrate a model orbit from the measured phase-space position, along which the stellar energy remains constant. If $\tau_{p,k}$ and $\tau_{p,k+1}$ are successive pericentric passages of this frozen model orbit, we define

$$\widetilde{J}_{r,\mathrm{BFE}}(t_*) \equiv \frac{1}{2\pi}\int_{\tau_{p,k}}^{\tau_{p,k+1}} p_r(\tau)\,\dot{r}(\tau)\,d\tau = \frac{1}{2\pi}\int_{\text{radial cycle}} p_r\,dr. \tag{36}$$

In an integrable spherical potential this reduces to the canonical radial action. In the frozen, non-spherical merger potential, however, one radial oscillation need not be a fundamental cycle of an invariant three-torus. The quantity may depend on phase, cycle number, and radial-cycle definition. We therefore refer to it as a frozen-potential radial-cycle action proxy rather than a true canonical action.

This proxy does not supply the theoretical basis of the KLD fit and is not an observable input to the method. It is used only as a simulation diagnostic of disagreement between the radial-cycle structure generated by the fitted spherical model and that generated by the frozen potential field.

3. SELECTING COHORTS OF STREAM STARS

The theoretical objective of the cohort selection is to retain stream material for which the instantaneous reference-action cloud is likely to have evolved approximately semi-rigidly through the time sequence of locally matched spherical reference potentials. For two cohort members a and b, this requires the total member-to-member difference

$$\Delta_{ab}\dot{J}_{0,i}(t) = \dot{J}_{0,i}^{(a)}(t) - \dot{J}_{0,i}^{(b)}(t) \tag{37}$$

to remain sufficiently small over the relevant evolutionary interval. Equivalently, the accumulated change

$$\Delta_{ab}J_{0,i}(t_0) = \Delta_{ab}J_{0,i}(t_1) + \int_{t_1}^{t_0} \Delta_{ab}\dot{J}_{0,i}(t)\,dt \tag{38}$$

should not strongly enlarge the pairwise action separations.

According to Equation 33, differential instantaneous-action evolution may arise through two channels. First, different cohort members may experience different residual forces relative to the locally matched spherical reference potential. Second, the members may respond differently when that reference potential evolves, because their assigned actions have different sensitivities to its changing parameters.

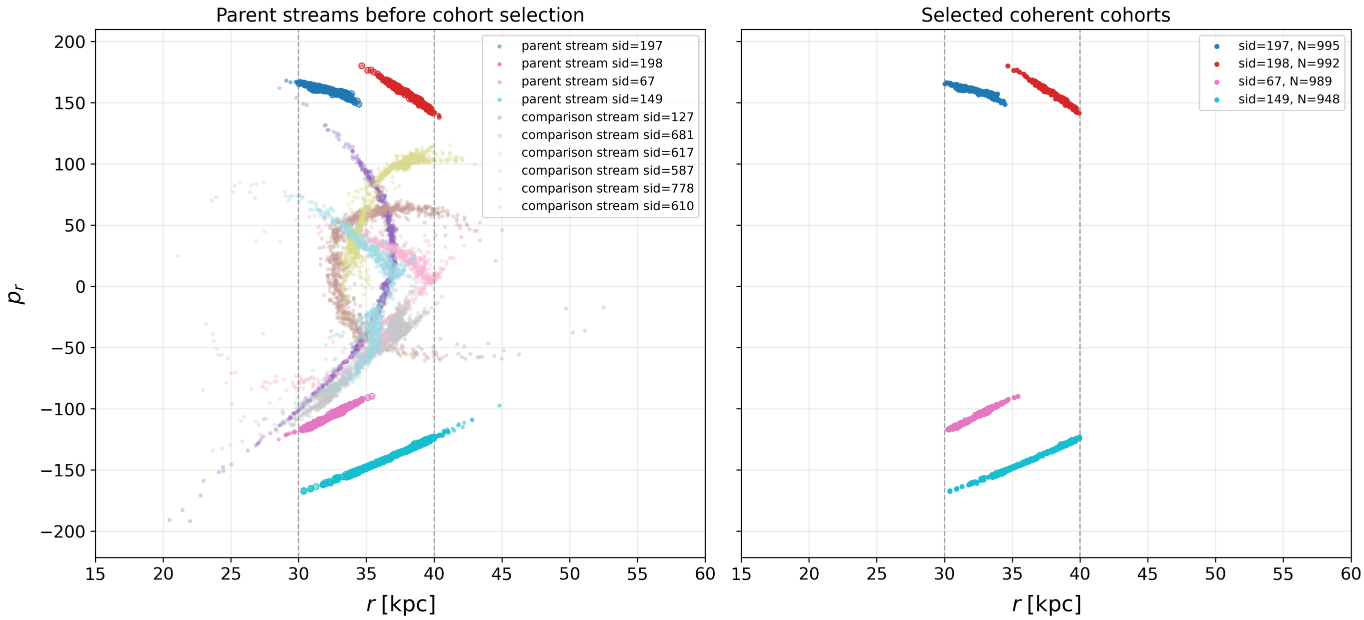

Figure 1. Illustration of the cohort-selection procedure in radial phase space for the 1:10 merger at snapshot 030 in the $30 \leq r/\mathrm{kpc} \leq 40$ radial range. The vertical dashed lines mark the boundaries of the selected radial interval. **Left:** four parent streams together with six comparison streams before cohort selection, including extended, folded, and kinematically irregular phase-space structures. **Right:** the locally coherent segments retained from the four parent streams after requiring a smooth along-stream velocity profile, an unfolded monotonic radial phase-space branch, limited spatial extent, and separation from radial turning points. The listed values of N give the stellar membership of each retained cohort. These criteria favor segments with smaller member-to-member differential forcing and reference-coordinate remapping, but do not by themselves guarantee an unbiased potential fit.

Neither the true force history nor the full sequence $\Phi_0(t)$ is available in an observation. We therefore construct the cohort selection using only present-day phase-space information. The criteria are designed to favor stream segments whose current morphology and kinematics are expected to accompany small member-to-member differences in the two contributions to $\dot{J}_0$. They are applied before the KLD potential fit to filter out less-informative observed materials.

First, we require a smooth velocity profile along the stream. If the cohort members experience similar residual forces relative to the evolving local spherical reference, their accumulated velocity responses should also remain similar. For the component parallel to the stream,

$$\delta v_{\parallel,a}(t_1 \to t_0) = \int_{t_1}^{t_0} \delta F_{\parallel,a}(t)\, dt + \text{const.} \tag{39}$$

A smooth present-day velocity profile is therefore expected when the member-dependent part of this accumulated response is small.

To implement this criterion, we introduce an along-stream coordinate s and fit a quadratic trend to the along-stream velocity:

$$v_{\rm fit}(s) = As^2 + Bs + C. \tag{40}$$

We retain only cohorts satisfying

$$\Delta_v \equiv \frac{\sqrt{\left\langle [v_\parallel(s) - v_{\rm fit}(s)]^2 \right\rangle}}{\mathrm{std}[v_\parallel(s)]} < 0.1. \tag{41}$$

Second, we require an unfolded, single-valued phase-space distribution. Members occupying the same smooth phase-space branch are more likely to have similar reference-orbit tangents,

$$\frac{\partial \boldsymbol{x}}{\partial \theta_{0,i}}, \tag{42}$$

and similar sensitivities of their instantaneous actions to the evolving reference potential,

$$\left. \frac{\partial J_{0,i}}{\partial \lambda_0^\alpha} \right|_{\boldsymbol{z}}. \tag{43}$$

These are the quantities multiplying the residual force and the reference-potential evolution in Equation 32. Stars belonging to different wrapped branches may occupy similar spatial locations while having substantially different velocities, radial phases, reference-orbit tangents, and action sensitivities. Their instantaneous actions can therefore respond differently even when they are evaluated under the same sequence $\Phi_0(t)$.

Operationally, we require the fitted velocity trend $v_{\rm fit}(s)$ to remain strictly monotonic over the selected interval. This removes strongly folded structures in which nearby positions correspond to different orbital branches.

Third, we require the selected cohort to have a limited spatial extent. A short segment occupies a comparatively restricted region of the true force field. It is therefore more likely that its members experience similar residual forces, $\delta \boldsymbol{F}$, and that one cohort-common locally matched spherical reference potential provides a meaningful description of their shared radial dynamics. Limited spatial extent also reduces member-to-member variation in both the reference-orbit tangent and the action sensitivity appearing in Equation 32.

At the same time, an extremely narrow radial segment may contain insufficient radial-phase and force-profile information to distinguish among trial spherical potentials. We therefore impose the minimum radial-span requirement

$$\Delta r_{\rm cohort} \equiv \max_a(r_a) - \min_a(r_a) \geq 2 \text{ kpc}. \tag{44}$$

The upper spatial-extent restriction favors coherent local dynamics, whereas the lower radial-span restriction preserves potential-identification leverage. In the present-day sample analyzed here, all candidate cohorts already exceed the 2 kpc lower limit, so this condition removes no additional present-day cohorts.

Fourth, we exclude cohorts lying close to radial turning points. Near pericentre or apocentre, v_r approaches zero and the mapping between local phase-space coordinates and radial phase becomes poorly conditioned. Stars at similar radii may lie on different incoming and outgoing branches and can consequently have different reference-orbit tangents and different sensitivities to the evolving reference potential. We therefore impose a threshold on $|v_r|$ to retain cohorts sufficiently far from radial turning points. This requirement is not redundant with the unfoldedness criterion, because a branch may remain single-valued while lying close to a turning point.

The resulting selection criteria are therefore:

1. a smooth along-stream velocity profile satisfying Equation 41;
2. an unfolded phase-space distribution, enforced by the monotonicity of $v_{\rm fit}(s)$;
3. a limited spatial extent together with the minimum radial span in Equation 44; and
4. sufficient separation from radial turning points, enforced by a threshold on $|v_r|$.

Together, these criteria favor locally coherent segments whose members are expected to experience similar residual-force responses and similar action-coordinate changes as the locally matched spherical reference potential evolves. In the terminology of Section 2.3, they are designed to preserve cohorts for which the instantaneous reference-action cloud can undergo approximately semi-rigid translation rather than strong internal diffusion.

The radial-phase-space comparison illustrates that this selection is deliberately restrictive. Before selection, the parent streams include extended, folded, and kinematically irregular structures, as well as material close to radial turning points. After applying the four criteria, the retained cohorts occupy shorter and more nearly single-valued phase-space branches. These are the cohorts for which the subsequent KLD maximization is more likely to distinguish trial-potential mismatch from action-cloud diffusion already produced by the merger.

The selection does not define the KLD-fitted potential and does not assume that the KLD result is already equal to Φ_0. Instead, it attempts to establish the dynamical precondition under which maximizing the clustering of trial-model actions can recover the locally appropriate spherical reference potential introduced in Section 2.3.

Fig. 1 illustrates how strongly these criteria reduce the parent-stream population to a small set of locally coherent segments. Before selection, the radial phase-space distribution contains extended streams with substantial curvature, overlap, and folded structures, including material close to orbital turning points. After applying the four selection criteria, the retained cohorts occupy much shorter and more nearly single-valued branches in radial phase space. The selected segments therefore exclude much of the spatially extended, kinematically irregular, and strongly phase-mixed material that would otherwise broaden the inferred action distribution.

This comparison also emphasizes that the selection is deliberately restrictive rather than simply removing a small number of outlying stars. The resulting cohorts represent only those local portions of the streams whose present-day phase-space structure provides strong evidence for coherent evolution history.

4. COMPUTATIONAL WORKS

4.1. *Simulation of merging halos with stellar streams*

We use the stellar-stream simulations from *Dancing Streams in Merging Halos* (Weerasooriya et al. 2025), a controlled suite designed to study how globular-cluster streams respond to a MW–LMC-like merger. The simulations first generate 1024 mock globular-cluster streams in a static MW-like Hernquist halo using the Fardal et al. (2015) mock-stream distribution function as implemented in Gala. Each stream contains 1000 stellar particles. The stream grid spans two different stellar masses, 10^4 and 10^6 $M_\odot$, eight apocenter radii from 30 to 100 kpc, eight circularities from $\eta = 0.5$ to 1.0, and eight stream ages from 0.5 to 6 Gyr, giving 1024 streams in total.

After generation, the same stream population is inserted into live N-body halos and evolved with Gadget4. The host halo has mass $M_{halo} = 1.57 \times 10^{12}$ $M_\odot$ and

Hernquist scale radius $a = 40.85$ *kpc*. The published *Dancing Streams in Merging Halos* suite describes the evolution of the streams in an isolated host and in a 1:5 MW–LMC-like merger, including fiducial, more circular, and radial perturber orbits. In this work, I also use additional related merger data products that extend the stream kinematics to a wider range of perturber-to-host mass ratios, 1:1, 1:2, 1:5, 1:10, and 1:100, with circular and radial perturber-orbit variants available for most ratios. This extended set allows me to study separately the effects of merger strength and perturber orbital geometry on the survival of coherent stream cohorts and on the bias of KLD-based spherical-NFW potential fits.. The full simulations are evolved for 6 Gyr with 241 snapshots. In this work, we analyze snapshots 000–100, corresponding to 0–2.56 Gyr. Our main KLD-fitting analysis focuses on selected cohorts in fixed present-day radial ranges, especially 20–30 kpc, with additional comparisons in 30–40 kpc and 50–70 kpc.

The potential model of the merger halo data packs can be described by the Basis Function Expansion (BFE), which expresses the mass density distribution as a superposition of basis functions:

$$\rho(\boldsymbol{x}, t) \approx \sum_{n,\ell,m} a_{n,\ell,m}(t)\, \rho_{n\ell m}(\boldsymbol{x}). \tag{45}$$

, where $\rho_{n,lm,}(\vec{x})$ is the basis function that consists of both the radial distribution and the angular distribution:

$$\rho_{n\ell m}(\boldsymbol{x}) = u_{n\ell}(r)\, Y_{\ell m}(\theta, \phi). \tag{46}$$

The relationship between the mass density distribution and the potential distribution can be described by Poisson's Equation:

$$\nabla^2 \phi(r) = 4\pi G \rho(r) \tag{47}$$

For those coefficients $a_{n,l,m}(t)$, I adopted the results worked out by David Amir Gonzalez, who is a graduate student at the University of Pennsylvania currently working with Professor Robyn Sanderson, upon their agreement. Their result accurately represent the later-stage potential structure in the simulation.

4.2. *Present-day recovery and time-dependent inner-halo performance*

4.2.1. *Implementing the Selection Criteria*

To test those selection criteria, I utilize them to pick coherent cohorts from the tidal streams simulated under an LMC-like merger, which has 1/10 of the mass of the Milky Way halo, merging into the Milky Way. The Milky Way halo is assumed to have a potential described by a spherical NFW model initially. Because the LMC is currently located at a Galactocentric distance of approximately 49.5 kpc (X. Wu et al. 2008) and is likely on its first infall or first passage about the Milky Way (G. Besla et al. 2007; N. Kallivayalil et al. 2013), we identify snapshot 55 ($t = 1.406$ Gyr), at which the simulated host–perturber separation is closest to the observed present-day MW–LMC separation, as the distance-matched present-day analogue used to examine the cohort-selection criteria.

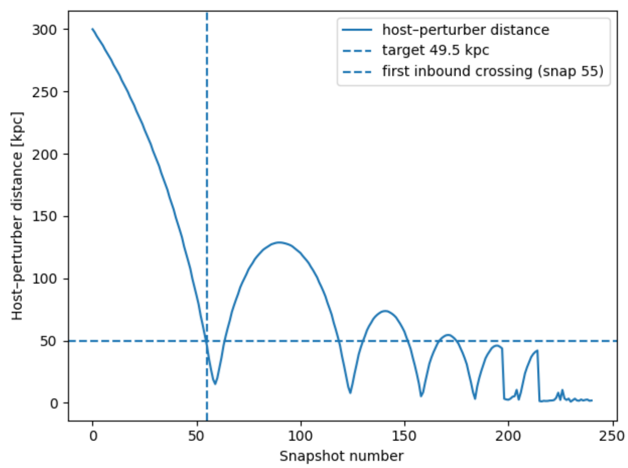

Figure 2. Host–perturber separation as a function of snapshot number for the fiducial Milky Way–LMC-like merger. The horizontal dashed line marks the target separation of 49.5 kpc, and the vertical dashed line identifies its first inbound crossing at snapshot 055 ($t = 1.406$ Gyr). We adopt this snapshot as the distance-matched present-day analogue used for the single-snapshot cohort-selection and enclosed–mass analysis.

In general, the gravitational potential of a perturbed, triaxial MW halo should be described by a general expansion, like the Basis Function Expansion (BFE). However, since I use data from a simulation that assumes a spherical potential initially, and that fitting a general BFE requires a much longer running time along with more stellar members to constrain the parameters, I followed the work of Brooks et al and optimized the parameters of a spherical NFW potential model.

Using those criteria, I selected coherent cohorts, each with at least 100 stars, in every 10-kpc-wide radial range from the galactic center outward. Each coherent cohort will give an optimized NFW model, using which I can integrate for an enclosed mass profile from the galactic center outward. One thing worth noting is that the perturbation force varies so strongly in the region near the center that no coherent cohort can be picked in the region of $r < 10$ kpc.

The spherical enclosed mass implied by an arbitrary gravitational potential can be obtained from the radial gravitational acceleration on a spherical surface. The gravitational acceleration is

$$\boldsymbol{g}(\boldsymbol{x}, t) = -\boldsymbol{\nabla}\Phi(\boldsymbol{x}, t), \tag{48}$$

while Poisson's equation gives

$$\nabla^2\Phi(\boldsymbol{x},t) = 4\pi G\rho(\boldsymbol{x},t). \tag{49}$$

Combining Equations 48 and 49, we obtain

$$\boldsymbol{\nabla}\cdot\boldsymbol{g}(\boldsymbol{x},t) = -4\pi G\rho(\boldsymbol{x},t). \tag{50}$$

We integrate Equation 50 over the volume V_r enclosed by a sphere of radius r:

$$\begin{aligned}\int_{V_r}\boldsymbol{\nabla}\cdot\boldsymbol{g}\,d^3\boldsymbol{x} &= -4\pi G\int_{V_r}\rho(\boldsymbol{x},t)\,d^3\boldsymbol{x}\\ &= -4\pi G M_{\rm sph}(<r,t).\end{aligned} \tag{51}$$

Applying the divergence theorem gives

$$\oint_{S_r}\boldsymbol{g}\cdot d\boldsymbol{S} = -4\pi G M_{\rm sph}(<r,t), \tag{52}$$

where S_r is the spherical surface bounding V_r.

On a sphere of radius r, the outward surface element is

$$d\boldsymbol{S} = r^2\widehat{\boldsymbol{r}}\,d\Omega, \tag{53}$$

and the radial component of the gravitational acceleration is

$$g_r(r,\Omega,t) \equiv \boldsymbol{g}(r,\Omega,t)\cdot\widehat{\boldsymbol{r}}. \tag{54}$$

Also, we can define the angular average of the radial acceleration at fixed radius as

$$\langle g_r(r,\Omega,t)\rangle_\Omega \equiv \frac{1}{4\pi}\int_{4\pi} g_r(r,\Omega,t)\,d\Omega. \tag{55}$$

Substituting Equations 53, 54 and 55 into Equation 52 yields the spherically averaged enclosed mass:

$$M_{\rm sph}(<r,t) = -\frac{r^2}{G}\left\langle g_r(r,\Omega,t)\right\rangle_\Omega. \tag{56}$$

The above equation does not require the gravitational potential to be spherical. It follows from the total gravitational flux through a spherical surface and therefore returns the mass enclosed by that surface implied by the potential model.

In my numerical calculation, the angular average in Equation 56 is evaluated numerically using $N_\Omega = 4096$ nearly uniformly distributed Fibonacci-sphere directions. Thus,

$$\langle g_r(r,\Omega,t)\rangle_\Omega \simeq \frac{1}{N_\Omega}\sum_{a=1}^{N_\Omega} g_r(r,\Omega_a,t), \tag{57}$$

and the enclosed mass is estimated as

$$M_{\rm sph}(<r,t) \simeq -\frac{r^2}{GN_\Omega}\sum_{a=1}^{N_\Omega} g_r(r,\Omega_a,t). \tag{58}$$

The Fibonacci construction provides nearly equal-area sampling over the sphere and is sufficiently dense relative to the angular resolution of the NFW spherical model.

In order to improve the precision, I decided to look at the weight-average enclosed-mass profile. In the left panel of Fig.3, I plotted the enclosed mass profile from each cohort only in the radial range where the cohort is located. Then I chopped the whole radial range into 0.1-kpc-wide bins. At each bin r_k, I weight-averaged over the enclosed mass values fitted from each cohort i that covers this bin:

$$M_{\rm fit}(<r_k,t) = \frac{\sum_{i\in\mathcal{I}_k} N_i M_{{\rm sph},i}(<r_k,t)}{\sum_{i\in\mathcal{I}_k} N_i}, \tag{59}$$

where $\mathcal{I}_k$ denotes the selected cohorts whose fitted profiles are evaluated at r_k, and N_i is the number of selected stars in cohort i. The right panel of Fig. 3 shows the weight-averaged enclosed mass value at each radial bin from the galactic center outward.

From this result, we can see that, despite the error caused by the model simplification, the streams selected by those criteria retrieve the enclosed mass profile reasonably precisely, especially in the range of $r \in [10, 30]$ kpc. Outside this radial range, the discrepancy arises as we go further out, which is expected. At snapshot 55, the center of mass of the merger is about 49.5 kpc away from the host halo. At such a radial distance or outward, a stream segment is more likely to survive as a coherent cohort if it lives in an area farther away from the perturber, where the local Dark Matter particles are more dominant by those from the host halo. Using the fitted local spherical potential at those lower-density and less-perturbed region to retrieve the spherically-averaged enclosed mass would naturally underestimate the global enclosed mass. So at the current merger stage of the LMC merger, the KLD-maximization method based on those selection criteria discussed above would work well to constrain the potential model inside the radial range sampled by the merger particles.

4.2.2. *Time-dependent inner-halo performance*

The distance-matched snapshot demonstrates that the selected cohorts can recover a useful enclosed-mass profile at a particular stage of the merger. However, this single-snapshot result does not establish how long the method remains reliable as the halo and stream population evolve. We therefore repeat the cohort selection and spherical-NFW fitting at every snapshot from 0 to 100, corresponding to $0 \le t \lesssim 2.56$ Gyr.

We first consider cohorts occupying the radial interval $20 \le r/\mathrm{kpc} \le 30$. For this inner-halo analysis, the reference enclosed mass is measured directly from the host and perturber dark-matter particles, rather than from the BFE potential.

To characterize the surviving stream sample independently of the tested potential, we define the six-dimensional phase-space compactness of cohort i as

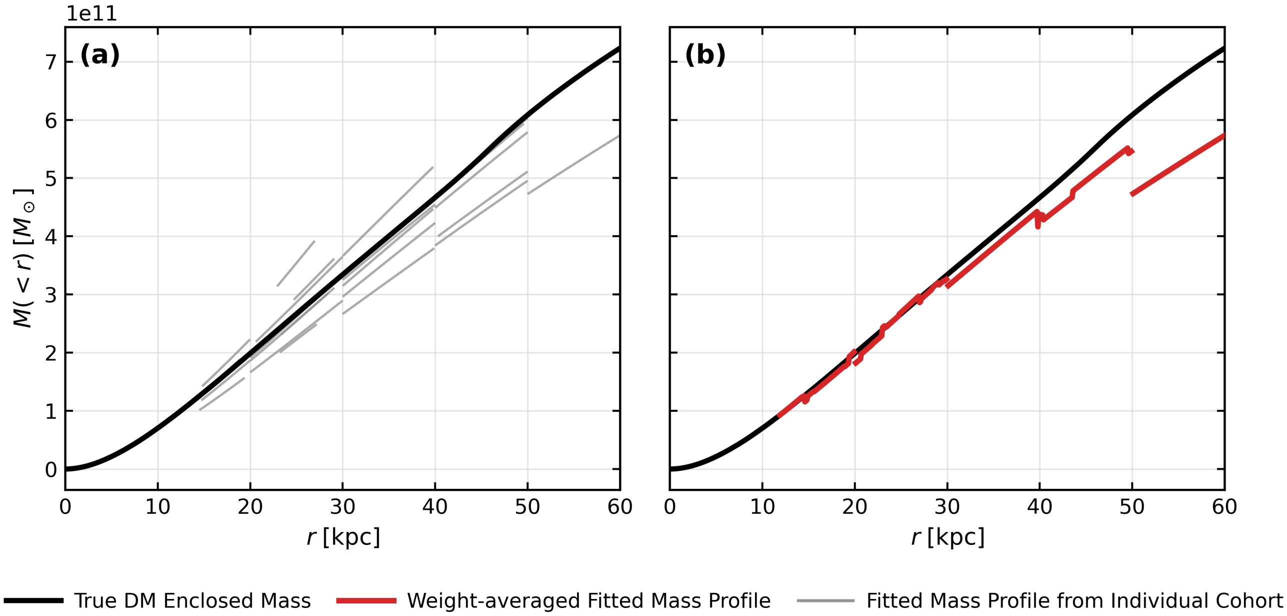

Figure 3. Enclosed-mass recovery from the selected coherent cohorts at snapshot 055, corresponding to the distance-matched present-day configuration of the fiducial 1:10 MW–LMC-like merger. **(a)** KLD-maximized spherical NFW enclosed-mass profiles inferred independently from the individual selected cohorts. Each grey curve is shown only over the radial interval sampled by its corresponding cohort, while the black curve shows the true dark-matter enclosed-mass profile measured directly from the snapshot particles relative to the host center. **(b)** Member-number-weighted mean of the independently fitted cohort mass profiles at each radius (red), compared with the same true enclosed-mass profile (black). Only cohorts whose sampled radial interval includes the given radius contribute to the weighted mean. Thus, the red curve is a radial combination of cohort-specific fits, rather than the result of a single joint fit to all selected stars. The recovered profile agrees most closely with the true profile at smaller radii, while the discrepancy generally increases farther outward. This behavior is consistent with the increasing importance of merger-induced departures from a spherical equilibrium model, together with the decreasing availability and radial coverage of useful coherent cohorts at large radii.

$$C_i = \left[\frac{1}{N_i}\sum_{j=1}^{N_i}\|\boldsymbol{z}_{ij} - \overline{\boldsymbol{z}}_i\|^2\right]^{1/2}, \tag{60}$$

where j is the star index, N_i is the number of selected stars in cohort i, and

$$\boldsymbol{z} = \left(\frac{x}{10\ \mathrm{kpc}}, \frac{y}{10\ \mathrm{kpc}}, \frac{z}{10\ \mathrm{kpc}}, \frac{v_x}{100\ \mathrm{km\,s^{-1}}}, \frac{v_y}{100\ \mathrm{km\,s^{-1}}}, \frac{v_z}{100\ \mathrm{km\,s^{-1}}}\right). \tag{61}$$

The selected-cohort-weighted compactness is

$$\langle C\rangle_w = \frac{\sum_i N_i C_i}{\sum_i N_i}. \tag{62}$$

At each radial grid point r_k within the 10-kpc-wide radial bin, the weighted fitted enclosed mass is

$$M_{\mathrm{fit}}(<r_k, t) = \frac{\sum_{i\in\mathcal{I}_k} N_i M_{\mathrm{sph},i}(<r_k, t)}{\sum_{i\in\mathcal{I}_k} N_i}, \tag{63}$$

where $\mathcal{I}_k$ denotes the selected cohorts whose fitted profiles are evaluated at r_k. We quantify the signed enclosed-mass bias by

$$b_M(t) = \left\langle \frac{M_{\mathrm{fit}}(<r_k, t) - M_{\mathrm{true}}(<r_k, t)}{M_{\mathrm{true}}(<r_k, t)} \right\rangle_k \times 100\%. \tag{64}$$

Figure 4 summarizes the resulting evolution calculated under the fiducial 1:10 merger. The first column shows the number of selected cohorts and their mean stellar membership, the second column shows $\langle C\rangle_w$, and the third column shows b_M.The distance-matched present-day reference time is marked by a light-red vertical line. The compactness generally increases with time, although it does not exhibit a single threshold that separates successful and unsuccessful fits. In section 4.3, we will discuss another set of quantities that are indeed related to the fitting precision. So far, the clearest merger-dependent hierarchy instead appears in the survival of the selected sample and in the reliability of the recovered mass.

The 1:2 merger loses most of its usable cohorts earliest and develops large, rapidly varying mass biases. The 1:5 and 1:10 cases retain selected material for longer, but subsequently exhibit substantial mass fluctuations. The 1:100 merger remains comparatively stable over most of

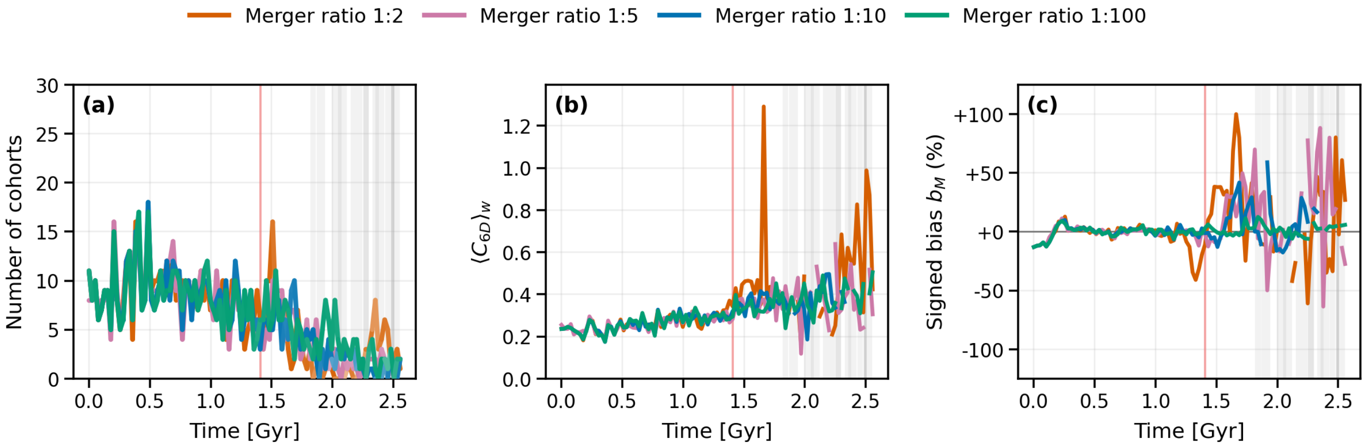

Figure 4. Time-dependent performance of the selected-cohort KLD fitting for merger mass ratios 1:2, 1:5, 1:10, and 1:100 in the $30 \leq r/\mathrm{kpc} \leq 40$ radial range. The light-red vertical line marks the distance-matched present-day reference time at snapshot 055 ($t = 1.406\,\mathrm{Gyr}$), defined using the fiducial 1:10 merger. **(a)** Number of selected coherent cohorts. Curve color identifies the merger ratio, while the local color saturation encodes the average stellar membership of the selected cohorts; more saturated segments correspond to larger average membership. **(b)** Cohort-member-number-weighted mean six-dimensional phase-space compactness, $\langle C_{6D}\rangle_w$. **(c)** Signed enclosed-mass bias, b_M, of the cohort-member-number-weighted fitted mass profile relative to the true dark-matter enclosed-mass profile measured directly from the snapshot particles and averaged over the radial grid points in the selected range. The background is white when all four merger cases retain at least one selected cohort and becomes progressively darker grey as fewer cases retain usable cohorts. Stronger mergers generally remove usable coherent stream material earlier and produce larger and more rapidly varying mass biases, whereas the 1:100 merger remains comparatively stable over most of the analyzed interval. The phase-space compactness evolves with time but does not by itself define a universal threshold for successful potential recovery.

the analyzed interval. Thus, stronger mergers generally shorten the interval over which a sufficiently large and dynamically coherent stream sample remains available and over which the fitted mass remains precise.

Occasional late-time snapshots may nevertheless return a small value of b_M. Such isolated agreement does not by itself demonstrate restored reliability. Because the cohorts are selected independently at each snapshot, the late-time sample can be dominated by an atypical subset of cohorts free of those moderately-perturbed yet passable members. We therefore interpret the time limit of the method from sustained behavior over multiple snapshots, rather than from individual near-zero mass residuals.

These inner-halo results establish the phenomenology of failure, but they do not distinguish why an available cohort produces an incorrect enclosed-mass estimate. We address this question in Section 4.3 using streams in the outer radial range $80 \leq r/\mathrm{kpc} \leq 100$, restricted to snapshot intervals over which the available BFE representation provides a sufficiently reliable description of the simulated force field.

For each selected cohort, we compare the canonical radial-action distribution calculated in the KLD-optimized spherical reference potential with the frozen-BFE radial-cycle action-proxy distribution defined in Equation 36. This comparison diagnoses disagreement between the radial-cycle mappings generated by the optimized spherical model and by the frozen BFE potential. We combine this diagnostic with a decomposition of the enclosed-mass bias into fitted-to-local and local-to-global contributions.

4.3. *Three failure modes of the KLD potential fit*

At radial grid point r_k, the fitted enclosed mass remains the cohort-size-weighted quantity

$$M_{\mathrm{fit}}(< r_k, t) = \frac{\sum_i N_i M_{\mathrm{sph},i}(< r_k, t)}{\sum_i N_i}. \tag{65}$$

For the BFE reference, we define the local spherical-equivalent mass at direction Ω as

$$M_{\mathrm{local},\ \Omega}(< r_k, \Omega, t) = -\frac{r_k^2}{G} g_r(r_k, \Omega, t), \tag{66}$$

and evenly average this quantity over the spatial directions sampled by the selected material,

$$M_{\mathrm{local}}(< r_k, t) = \frac{1}{N_{\Omega,k}} \sum_{a=1}^{N_{\Omega,k}} M_{\mathrm{local},\ \Omega}(< r_k, \Omega_a, t), \tag{67}$$

where $N_{\Omega,k}$ is the number of spacial directions in the angular area sampled by the selected cohorts at r_k.

We therefore define

$$b_{M,\mathrm{fit}\to\mathrm{local}}(t) = 100\left\langle \frac{M_{\mathrm{fit}}(<r_k,t) - M_{\mathrm{local}}(<r_k,t)}{M_{\mathrm{true}}(<r_k,t)} \right\rangle_k, \tag{68}$$

and

$$b_{M,\mathrm{local}\to\mathrm{true}}(t) = 100\left\langle \frac{M_{\mathrm{local}}(<r_k,t) - M_{\mathrm{true}}(<r_k,t)}{M_{\mathrm{true}}(<r_k,t)} \right\rangle_k \tag{69}$$

in the radial range of $80 \leq r/\mathrm{kpc} \leq 100$. The radial-grid average $\langle\cdots\rangle_k$ is an ordinary equal-bin average. The first quantity measures whether the KLD-maximized spherical potential reproduces the effective local gravitational field sampled by the selected stream material, whereas the second quantity measures whether that local field is representative of the globally spherical-averaged BFE potential field. Thus, when a usable selected-cohort population is available, the total signed mass bias can be written as

$$b_M(t) = b_{M,\mathrm{fit}\to\mathrm{local}}(t) + b_{M,\mathrm{local}\to\mathrm{true}}(t). \tag{70}$$

In the following subsections, we will discuss about those two components more specifically.

4.3.1. *Failure mode I: loss of coherent stream material*

The first failure mode occurs when the merger stretches, folds, heats, or otherwise perturbs the streams strongly enough that few or no stellar segments satisfy the coherence criteria. In this regime, the limitation is not primarily that the KLD optimizer selects an incorrect potential; rather, the data required for the optimization are no longer available. Similar to the previous inner-range analysis, the fitted results for a 1:2 merger show discontinuity at earlier time, due to the earlier removal of usable coherent stream material.

4.3.2. *Mismatch between the fitted spherical-action and frozen-BFE proxy mappings*

A misspecified spherical potential may produce a compact radial-action distribution while placing that distribution at the wrong action-space location or producing a different degree of clustering. To isolate this failure mode from the separate local-to-global effect discussed in Section 4.3.3, we compare the action-cloud discrepancies specifically with $b_{M,\mathrm{fit}\to\mathrm{local}}$, rather than with the total mass bias b_M.

For every selected star, let $J_{r,\mathrm{fit}}$ denote the canonical radial action evaluated in the KLD-optimized spherical NFW potential, and let $\widetilde{J}_{r,\mathrm{BFE}}$ denote the frozen-BFE radial-cycle action proxy defined in Equation 36. We characterize the displacement between the two cohort distributions using

$$\Delta\widetilde{J}_{r,\mathrm{med}} \equiv \mathrm{median}(J_{r,\mathrm{fit}}) - \mathrm{median}\left(\widetilde{J}_{r,\mathrm{BFE}}\right), \tag{71}$$

and characterize their difference in clustering using

$$\Delta D_{\mathrm{KL},r} \equiv D_{\mathrm{KL}}[J_{r,\mathrm{fit}}] - D_{\mathrm{KL}}\left[\widetilde{J}_{r,\mathrm{BFE}}\right]. \tag{72}$$

The two KLD values are evaluated using the same fixed reference distribution and action interval.

We retain the signs of both quantities. The sign of $\Delta\widetilde{J}_{r,\mathrm{med}}$ records the direction of the displacement between the spherical-reference action cloud and the frozen-BFE radial-cycle proxy cloud. The sign of $\Delta D_{\mathrm{KL},r}$ indicates whether the optimized spherical model produces a more or less clustered distribution than the frozen-BFE radial-cycle mapping.

These quantities should not be interpreted as differences between two sets of exact actions. Only $J_{r,\mathrm{fit}}$ is a canonical radial action. The BFE quantity is a frozen-potential radial-cycle proxy. Their discrepancy is therefore a diagnostic of disagreement between the radial-cycle structures generated by the two potentials, rather than a direct metric of the potential difference itself.

For every selected star a in cohort i at snapshot t, let $J_{r,\mathrm{fit},ia}(t)$ denote the canonical radial action evaluated in the KLD-optimized spherical NFW potential, and let $\widetilde{J}_{r,\mathrm{BFE},ia}(t)$ denote the frozen-BFE radial-cycle action proxy defined in Equation 36. Before constructing the action-discrepancy diagnostics, we match the fitted and BFE samples by stellar ID and retain only stars for which both quantities are finite. We denote the number of these matched stars in cohort i by $N_i^{\mathrm{match}}(t)$.

The signed displacement of the median radial coordinate is first calculated separately within each cohort:

$$\Delta\widetilde{J}_{r,\mathrm{med},i}(t) \equiv \mathrm{median}_{a\in i}\left[J_{r,\mathrm{fit},ia}(t)\right] - \mathrm{median}_{a\in i}\left[\widetilde{J}_{r,\mathrm{BFE},ia}(t)\right]. \tag{73}$$

A positive value therefore means that the fitted spherical model assigns a larger median radial action than the frozen-BFE radial-cycle mapping.

Let $p_{i,\mathrm{fit}}$ and $\widetilde{p}_{i,\mathrm{BFE}}$ denote, respectively, the distributions of $J_{r,\mathrm{fit}}$ and $\widetilde{J}_{r,\mathrm{BFE}}$ for the same matched cohort members. The corresponding signed difference in clustering is

$$\Delta D_{\mathrm{KL},r,i}(t) \equiv D_{\mathrm{KL}}[p_{i,\mathrm{fit}}\|q] - D_{\mathrm{KL}}[\widetilde{p}_{i,\mathrm{BFE}}\|q], \tag{74}$$

where both KLD values are evaluated using the same fixed reference distribution q and the same global action interval $[J_{r,\mathrm{min}}, J_{r,\mathrm{max}}]$. Thus, $\Delta D_{\mathrm{KL},r,i} > 0$ means that the fitted-model distribution is more strongly clustered than the frozen-BFE proxy distribution. The cohort-level KLD comparison is retained only when $N_i^{\mathrm{match}} \geq N_{\mathrm{min}} = 50$.

The quantities plotted in Figures 8 are obtained by first calculating Equations 73 and 74 within each cohort and then forming weighted means over the contributing

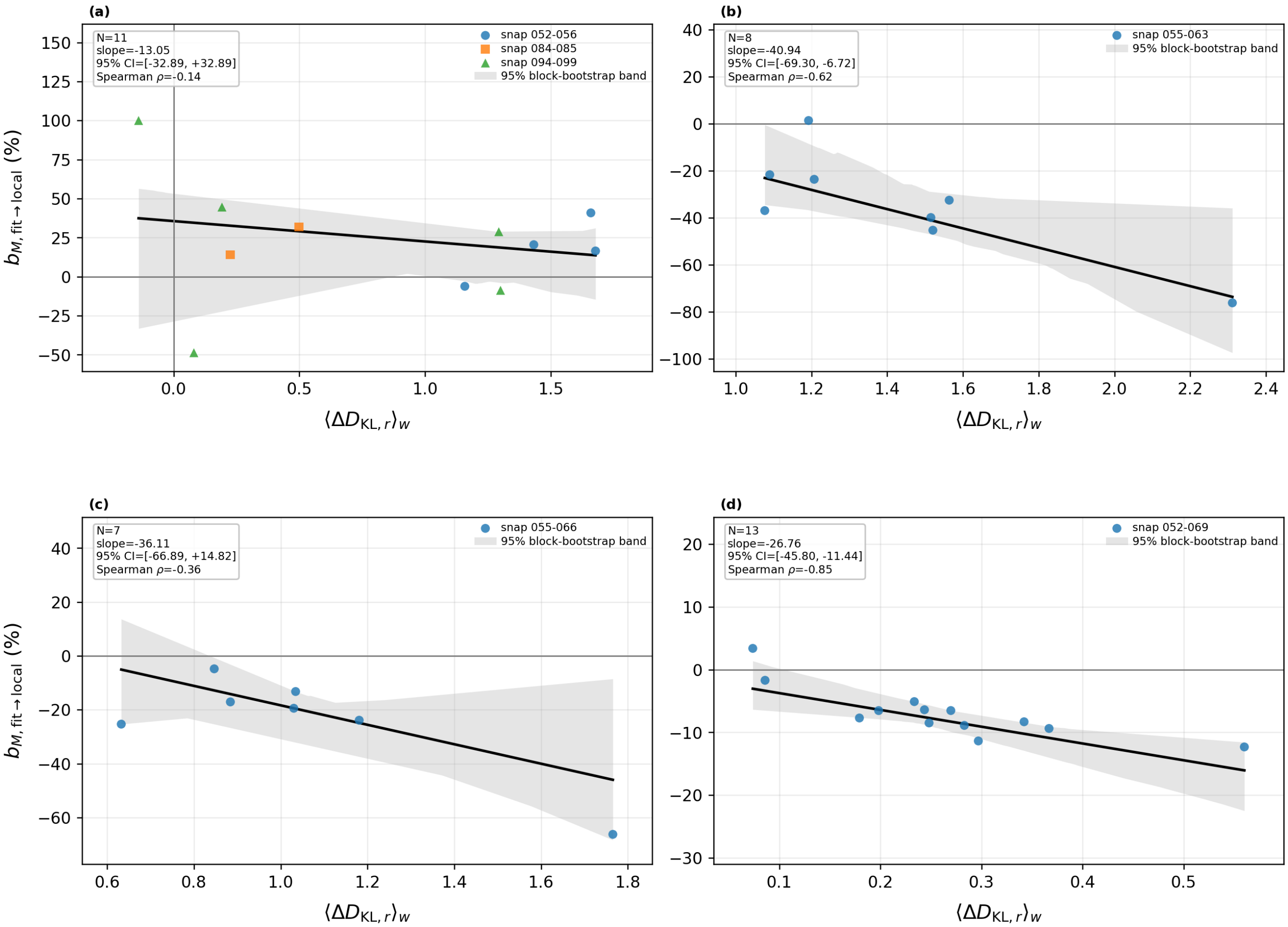

Figure 5. Fit-to-local enclosed-mass bias, $b_{M,\mathrm{fit}\to\mathrm{local}}$, versus the matched-star-count-weighted fitted-minus-BFE clustering difference, $\langle \Delta D_{\mathrm{KL},r} \rangle_w$, for the BFE-valid 80–100 kpc snapshot intervals. Panels **(a)**–**(d)** show the 1:2, 1:5, 1:10, and 1:100 mergers, respectively. Each point represents one snapshot, and different marker shapes distinguish disconnected contiguous BFE-valid intervals. Black lines show Theil–Sen robust linear fits, while the gray regions show the 95% moving-block-bootstrap confidence bands, which account approximately for temporal correlation between neighboring snapshots. The annotation boxes give the number of plotted snapshots, the fitted slope and its bootstrap interval, and the Spearman rank coefficient. These relations are descriptive diagnostics and should not be interpreted as universal linear scaling laws.

cohorts:

$$\left\langle \Delta \widetilde{J}_{r,\mathrm{med}} \right\rangle_w (t) \equiv \frac{\displaystyle\sum_{i \in \mathcal{C}_{\mathrm{med}}(t)} N_i^{\mathrm{match}}(t)\, \Delta \widetilde{J}_{r,\mathrm{med},i}(t)}{\displaystyle\sum_{i \in \mathcal{C}_{\mathrm{med}}(t)} N_i^{\mathrm{match}}(t)}, \tag{75}$$

and

$$\langle \Delta D_{\mathrm{KL},r} \rangle_w (t) \equiv \frac{\displaystyle\sum_{i \in \mathcal{C}_{\mathrm{KLD}}(t)} N_i^{\mathrm{match}}(t)\, \Delta D_{\mathrm{KL},r,i}(t)}{\displaystyle\sum_{i \in \mathcal{C}_{\mathrm{KLD}}(t)} N_i^{\mathrm{match}}(t)}. \tag{76}$$

Here, $\mathcal{C}_{\mathrm{med}}(t)$ and $\mathcal{C}_{\mathrm{KLD}}(t)$ are the sets of cohorts with finite median-displacement and KLD-discrepancy measurements, respectively. The latter additionally satisfies the minimum matched-star requirement. These definitions give greater statistical weight to better-sampled cohorts while preserving the cohort-by-cohort construction of the fitted potential and action distribution. If no cohort contributes a finite diagnostic at a snapshot, the corresponding weighted quantity is undefined and is displayed as a gap in the time series.

These diagnostics compare a canonical radial-action distribution in the fitted spherical model with a frozen-potential radial-cycle proxy distribution. They should therefore not be interpreted as differences between two exact actions or as direct measurements of the potential difference. Instead, they diagnose disagreement between the radial-cycle mappings generated by the fitted spherical model and by the frozen BFE reference field.

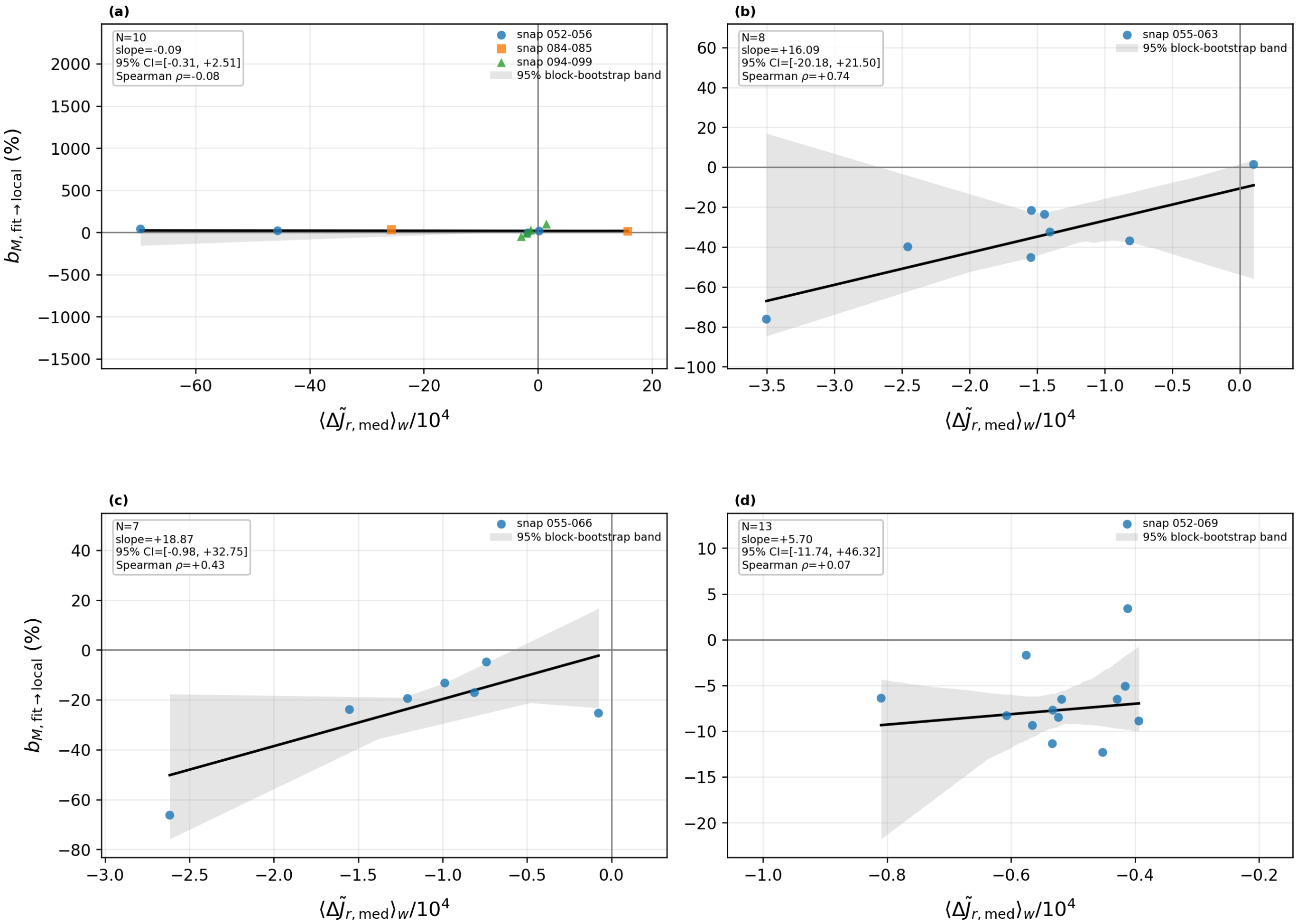

Figure 6. Fit-to-local enclosed-mass bias, $b_{M,\mathrm{fit}\to\mathrm{local}}$, versus the matched-star-count-weighted fitted-minus-BFE median displacement, $\langle \Delta \widetilde{J}_{r,\mathrm{med}} \rangle_w$, for the BFE-valid 80–100 kpc snapshot intervals. The radial-action displacement is scaled by 10^4 as indicated on the horizontal axes. Panels **(a)**–**(d)** show the 1:2, 1:5, 1:10, and 1:100 mergers, respectively. Each point represents one snapshot, and different marker shapes distinguish disconnected contiguous BFE-valid intervals. Black lines show Theil–Sen robust linear fits, and the gray regions show the 95% moving-block-bootstrap confidence bands. The annotation boxes report the number of plotted snapshots, the fitted slope and its bootstrap interval, and the Spearman rank coefficient. For panel **(a)**, snapshot 097, the most extreme median-displacement point in the snapshots 094–099 interval, is omitted from this scatter plot and its robust fit for visibility; it remains included in the full component table and in the time-series diagnostics. The fitted relations are descriptive rather than universal scaling laws.

Figure 5 and 6 shows these two signed action-cloud diagnostics against $b_{M,\mathrm{fit}\to\mathrm{local}}$ for the different merger strengths. We summarize each relation using a Theil–Sen robust linear estimator. Because adjacent snapshots within a BFE-valid interval are temporally correlated, the uncertainty in the fitted slope is estimated using a moving-block bootstrap within each contiguous interval. These regressions should be interpreted as descriptive diagnostics rather than universal linear scaling relations.

From these results, a larger $\langle \Delta D_{\mathrm{KL},r} \rangle_w$ tends to be associated with a more negative fit-to-local mass bias. One possible interpretation is that the spherical NFW fit partially compensates for merger-induced perturbations by selecting a potential that artificially compresses the inferred $J_{r,\mathrm{fit}}$ distribution relative to the frozen-BFE radial-cycle proxy distribution. This excess clustering appears to be associated with a shallower locally inferred force field and therefore an underestimated enclosed mass. Likewise, for the 1:5, 1:10, and 1:100 cases, a positive displacement of the fitted median radial action, $\langle \Delta \widetilde{J}_{r,\mathrm{med}} \rangle_w$, tends to accompany a more positive enclosed-mass bias. One possible interpretation of this positive relation is that a fitted NFW model with a larger enclosed mass produces a deeper effective potential and changes the reconstructed radial orbit so that the same observed phase-space coordinates enclose a larger radial phase-space area. The resulting increase in the fitted median $J_{r,\mathrm{fit}}$ would then accompany a positive local enclosed-mass bias.

The resulting trends vary strongly with merger strength. The 1:5 and 1:10 mergers show appreciable central relations between the weighted action-cloud discrepancies and $b_{M,\mathrm{fit}\to\mathrm{local}}$. For $\langle \Delta D_{\mathrm{KL},r} \rangle_w$, their fitted slopes are approximately -40.94 and -36.11, respectively, although the 95% bootstrap interval includes zero for the 1:10 case. For $\langle \Delta \widetilde{J}_{r,\mathrm{med}} \rangle_w / 10^4$, the corresponding slopes are approximately $+16.09$ and $+18.87$, respectively, but both bootstrap intervals include zero. The 1:100 system occupies a much narrower range in both action discrepancy and mass bias, as expected for the least strongly perturbed halo. Nevertheless, its KLD-based relation is strongly monotonic, with a slope of approximately -26.76 and a Spearman coefficient of $\rho = -0.85$, whereas its median-displacement relation is considerably weaker.

The 1:2 merger case is special. As shown in Figures 5 and 6, its fitted relations are comparatively weak: the slope is approximately -13.05 for the $\langle \Delta D_{\mathrm{KL},r} \rangle_w$ relation and -0.09 for the $\langle \Delta \widetilde{J}_{r,\mathrm{med}} \rangle_w / 10^4$ relation, with correspondingly weak rank correlations of $\rho = -0.14$ and $\rho = -0.08$. The respective 95% bootstrap intervals, $[-32.89, +32.89]$ and $[-0.31, +2.51]$, both include zero. The weak scalar relations in the 1:2 merger likely indicate a breakdown of a one-dimensional mapping between radial action-cloud mismatch and local enclosed-mass bias. In this strongly perturbed regime, the fitted canonical-action and frozen-BFE radial-cycle proxy distributions may differ in width, skewness, or multimodality in ways that are not captured by either $\langle \Delta D_{\mathrm{KL},r} \rangle_w$ or $\langle \Delta \widetilde{J}_{r,\mathrm{med}} \rangle_w / 10^4$. Moreover, the radial-action diagnostics depend on the force profile over the full reconstructed radial cycle, which may extend far inside and outside 80–100 kpc, whereas $b_{M,\mathrm{fit}\to\mathrm{local}}$ compares radial forces over the selected radial shell and directions. In the strongly perturbed 1:2 merger case, changes in the global radial profile or higher multipoles of the potential can strongly affect the radial-action distributions without producing a proportional change in the local spherical-equivalent mass. Rapid changes in the surviving cohort population and the combination of disconnected snapshot intervals may introduce additional scatter. Therefore, the shallow slopes do not necessarily indicate the absence of an action-space failure, but rather that this failure is inadequately represented by either scalar diagnostic alone.

4.3.3. *Failure mode III: local-to-global mismatch in a non-spherical potential*

Even if a spherical fitted potential accurately reproduced the gravitational field locally sampled by a selected cohort, the resulting spherical-equivalent enclosed mass would not necessarily equal the globally spherical-averaged enclosed mass.

For the full BFE force field, the globally spherical-averaged enclosed mass is

$$M_{\mathrm{true}}(< r, t) = -\frac{r^2}{G} \left\langle g_r(r, \Omega, t) \right\rangle_{\Omega} . \tag{77}$$

At an individual direction Ω, we define

$$M_{\mathrm{local},\ \Omega}(< r, \Omega, t) = -\frac{r^2}{G} g_r(r, \Omega, t). \tag{78}$$

This quantity is the spherical enclosed mass that would reproduce the true BFE radial acceleration at that location.

We evaluate $M_{\mathrm{local}\ \Omega}$ along the angular directions sampled by the selected outer-halo material and form the unweighted sampled mean M_{local}. Then the corresponding local-to-true contribution to the mass bias is

$$b_{M,\mathrm{local}\to\mathrm{true}}(t) = 100 \left\langle \frac{M_{\mathrm{local}}(< r_k, t) - M_{\mathrm{true}}(< r_k, t)}{M_{\mathrm{true}}(< r_k, t)} \right\rangle_k . \tag{79}$$

Figure 7 shows the time evolution of $b_{M,\mathrm{local}\to\mathrm{true}}$ during the same BFE-valid intervals used for the action-cloud comparison. The local-to-true mismatch is noticeably larger for the stronger mergers, but its absolute magnitude remains comparatively modest. Even in the 1:2 merger, $b_{M,\mathrm{local}\to\mathrm{true}}$ remains predominantly within approximately ten percent of zero over the sampled intervals. The 1:5 and 1:10 mergers typically show only several-percent offsets, while the 1:100 case remains still closer to zero.

Non-sphericity therefore contributes a genuine additional uncertainty to the interpretation of a locally fitted spherical potential, but the present decomposition indicates that it is *not the dominant source of the large mass failures* seen in the strongly perturbed cases. In particular, the tens-of-percent excursions in the fitted mass cannot generally be produced by a local-to-true term whose magnitude is mostly below ten percent. Most of the large discrepancy must instead arise in $b_{M,\mathrm{fit}\to\mathrm{local}}$, consistent with failure of the KLD-maximized spherical potential to recover the appropriate instantaneous local action structure.

The second local-to-true term in Equation 70 can nevertheless affect the detailed snapshot-to-snapshot total bias. Its sign can either reinforce or partially cancel the contribution of the first fit-to-local term, so it may introduce additional scatter when the total b_M is compared directly with a scalar action-cloud diagnostic. Its role should therefore be regarded as a *secondary non-sphericity correction*, rather than as the principal explanation for the breakdown of the KLD fit in the 1:2 merger.

4.3.4. *Combined reliability conditions*

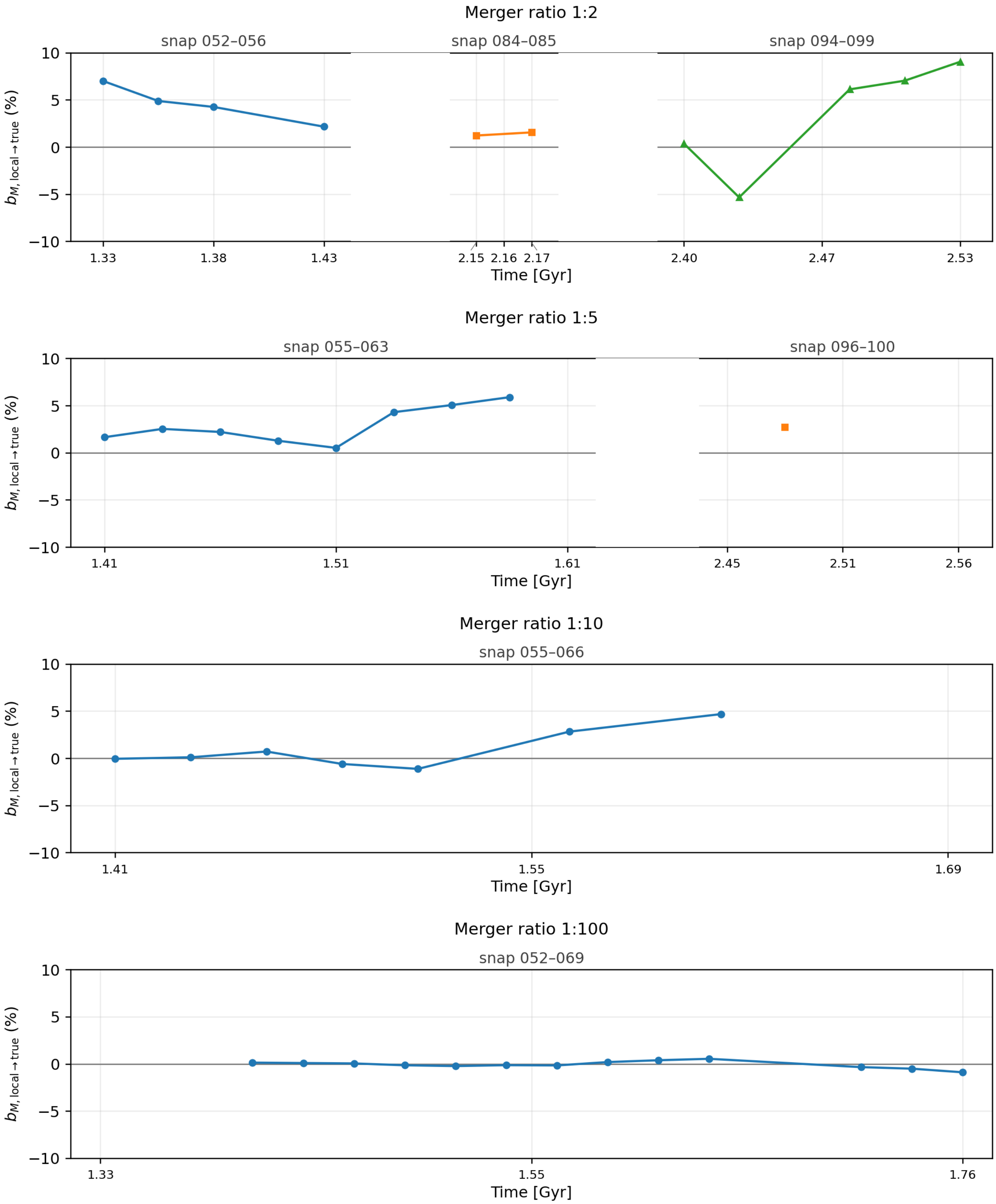

Figure 7. Local-to-global contribution to the enclosed-mass bias, $b_{M,\mathrm{local}\to\mathrm{true}}$, evaluated using the selected outer-range cohorts during the BFE-valid snapshot intervals. From top to bottom, the rows show the 1:2, 1:5, 1:10, and 1:100 mergers. At each radial grid point, the locally sampled spherical-equivalent mass is inferred from the BFE radial force over the angular region occupied by the selected cohorts and is compared with the globally spherically averaged BFE enclosed mass. Disconnected segments and different marker styles identify separate contiguous BFE-valid intervals. The mismatch remains mostly below 10% even for the 1:2 merger and is generally only a few percent for the weaker mergers. Thus, non-sphericity and non-uniform spatial sampling contribute to the total mass bias but are typically subdominant to the fit-to-local discrepancy.

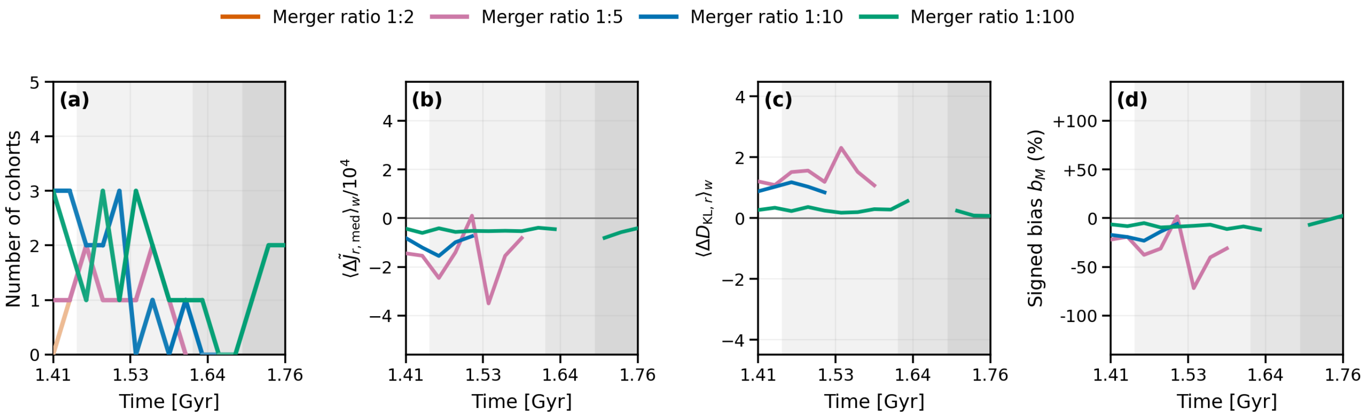

Figure 8. Comparison of the KLD-fitted spherical NFW model with the frozen-snapshot BFE reference field for merger mass ratios 1:2, 1:5, 1:10, and 1:100 in the $80 \leq r/\mathrm{kpc} \leq 100$ radial range. Only BFE-valid snapshot intervals within $1.41 \leq t/\mathrm{Gyr} \leq 1.76$ contribute to the comparison. **(a)** Number of selected coherent cohorts. The color identifies the merger ratio. The saturation of each curve segment encodes the average stellar membership of the selected cohorts, which typically ranges from 870 to 950; more saturated segments correspond to larger average membership. **(b)** Matched-star-count-weighted signed median displacement, $\langle \Delta \widetilde{J}_{r,\mathrm{med}} \rangle_w$, scaled by 10^4 as indicated on the axis. **(c)** Matched-star-count-weighted signed clustering difference, $\langle \Delta D_{\mathrm{KL},r} \rangle_w$. **(d)** Signed enclosed-mass bias, b_M, of the fitted spherical model relative to the globally spherically averaged BFE reference. The action diagnostics are defined in Equations 75 and 76. White background indicates times at which all four merger cases lie within BFE-valid intervals; progressively darker gray indicates that fewer cases provide a valid BFE comparison. Missing curve segments indicate that no valid diagnostic is available at that time.

The KLD method therefore has no single universal failure threshold. Reliable recovery requires the simultaneous satisfaction of three conditions:

1. a sufficiently large population of dynamically coherent stream cohorts must remain available;
2. the KLD-maximized fitted potential must reproduce the relevant location and clustering structure of the instantaneous action cloud under the true frozen-snapshot potential; and
3. the local gravitational field sampled by the retained cohorts must be sufficiently representative of the global potential distribution.

The new mass-bias decomposition shows that these three limitations are not equally important in the regimes considered here. When a merger becomes sufficiently strong that almost no coherent material survives, failure mode I dominates because the tracer population itself has disappeared. While usable cohorts remain, however, the largest mass errors are generally associated with failure mode II: the KLD-maximized spherical model does not reproduce the appropriate instantaneous BFE action-space structure. Failure mode III contributes an additional offset from the non-spherical spatial variation of the true force field, but this contribution remains mostly below the ten-percent level even in the 1:2 merger. Figure 8 provides a direct comparison between the evolution of the selected stream sample, the discrepancies between the cloud of fitted action versus that of BFE action-proxy, and the resulting enclosed-mass bias in the outer-halo radial range of $80 \leq r/\mathrm{kpc} \leq 100$.

5. DISCUSSION AND CONCLUSION

5.1. *Dependence on perturber orbital configuration*

Figure 9 compares the standard, circular, and radial 1:5 merger configurations over their respective BFE-valid intervals. The radial configuration appears most favorable for the KLD fitting method over the periods analyzed here. Although the close radial encounter may produce a strong transient disturbance, the strongest forcing is plausibly concentrated into a relatively short interval. Outside this interval, coherent cohorts remain or reappear, and the fitted-minus-BFE action discrepancies and enclosed-mass biases are comparatively modest over much of the sampled evolution.

The circular configuration is less favorable in terms of sustained tracer availability. One possible interpretation is that prolonged merger-induced forcing continuously distorts the halo and stream population. The surviving action clouds generally have positive $\Delta D_{\mathrm{KL},r}$ and negative $\Delta \widetilde{J}_{r,\mathrm{med}}$, indicating that the optimized spherical NFW model produces an action distribution that is more clustered and shifted toward lower radial actions relative to the BFE reference.

This orbital comparison should be regarded as suggestive rather than as a universal ranking of perturber trajectories, since the analysis is restricted to intervals in which BFE provides a sufficiently reliable reference. However, the comparison indicates that not only merger mass ratio, but also the duration and geometry of the

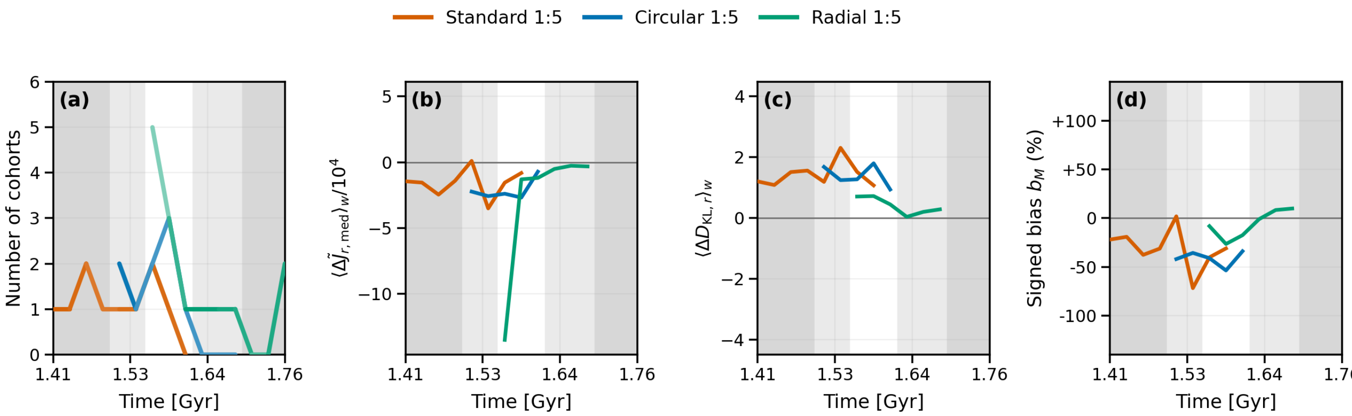

Figure 9. Comparison of the KLD-fitted spherical NFW model with the frozen-snapshot BFE reference field for the standard, circular, and radial 1:5 merger configurations in the $80 \leq r/\mathrm{kpc} \leq 100$ radial range. Only BFE-valid snapshot intervals within $1.41 \leq t/\mathrm{Gyr} \leq 1.76$ contribute to the comparison. **(a)** Number of selected coherent cohorts. Curve color identifies the orbital configuration, while increasing color saturation represents increasing average stellar membership per cohort. **(b)** Matched-star-count-weighted signed median displacement, $\langle \Delta \widetilde{J}_{r,\mathrm{med}} \rangle_w$, scaled by 10^4 as indicated on the axis. **(c)** Matched-star-count-weighted signed clustering difference, $\langle \Delta D_{\mathrm{KL},r} \rangle_w$. **(d)** Signed enclosed-mass bias, b_M, of the fitted spherical model relative to the globally spherically averaged BFE reference. White background indicates times at which all three configurations lie within BFE-valid intervals; progressively darker gray indicates that fewer configurations provide a valid BFE comparison. Missing curve segments indicate that no valid diagnostic is available.

encounter, affect the survival of coherent stream material and the form in which the action-space fitting failure appears.

5.2. *Limitations of observable cohort selection and future prospects*

The selection procedure introduced in Section 3 can effectively improve the KLD-maximization potential-constraining method by removing obviously unsuitable material. By retaining short, unfolded stream segments with smooth velocity profiles and locations away from orbital turning points, the method removes material that is most clearly phase-mixed or kinematically irregular. Nevertheless, it certainly does not provide a guarantee of unbiased potential recovery. The results of Section 4.3 demonstrate that apparently coherent cohorts may still yield biased fits when the optimized spherical potential does not reproduce the correct instantaneous action mapping or when the locally sampled gravitational field is not representative of the global spherical average. These residual errors cannot generally be removed using the present selection criteria alone.

This limitation is especially relevant observationally because the true time-dependent potential and its corresponding instantaneous action distribution are not known. The comparisons from direct Dark Matter particle number-counting and from BFE potential models we present are therefore validation diagnostics only available in simulations, rather than quantities that can be measured directly for Galactic streams. Future work should investigate whether the onset of failure can instead be identified through observable or internally testable signatures, such as disagreement among independently fitted cohorts, sensitivity of the inferred potential to cohort removal, inconsistency between different radial ranges, or disagreement among multiple action-space diagnostics.

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